E-Commerce Customer Segmentation

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Data Analytics Portfolio Project
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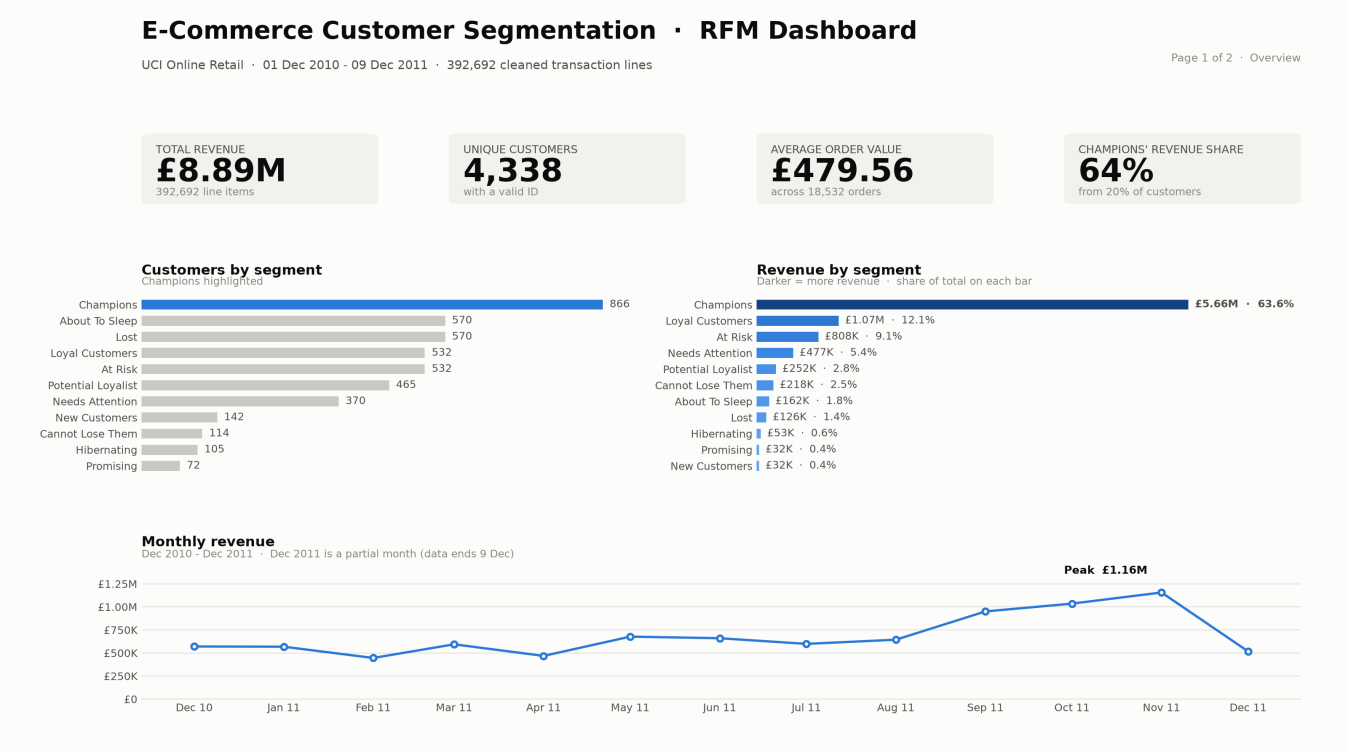
RFM (Recency, Frequency, Monetary) analysis of 541,909 transactions · Python · PostgreSQL · Power BI

Executive summary

A UK online retailer held 13 months of transaction history — 541,909 raw rows — but no view of which customers actually drove revenue, so marketing spend was spread evenly across the base. I built a reproducible pipeline that cleans the raw extract, scores every customer on Recency, Frequency and Monetary value using SQL window functions, and segments the base into eleven actionable groups.

The result reframes the budget conversation: **20% of customers generate 64% of revenue**, and a further 646 high-value customers are lapsing with £1.03M of historic spend at stake. Both groups are now addressable by name.

REVENUE ANALYSED	CUSTOMERS SCORED	AVERAGE ORDER VALUE	CHAMPIONS' SHARE
£8.89M	**4,338**	**£480**	**64%**
18,532 orders	11 segments	across 13 months	from 20% of customers



Dashboard page 1 — segment sizes, revenue distribution and monthly trend. Full two-page dashboard: reports/rfm_dashboard.pdf

What the analysis found

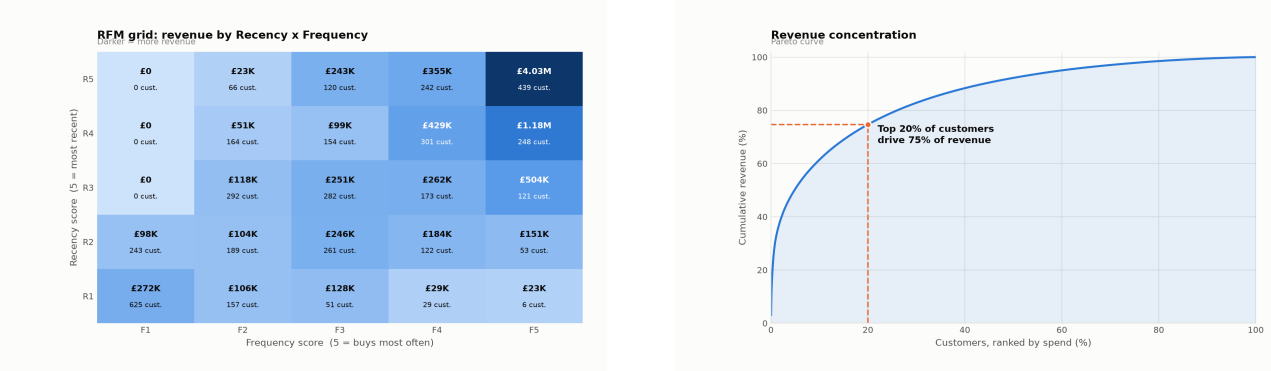
- **Revenue is highly concentrated.** The top 20% of customers by spend account for 75% of revenue; the bottom half contributes under 4%.
- **866 Champions are the business.** They buy recently, often, and at the highest value — £5.66M between them.
- **114 'Cannot Lose Them' customers have gone quiet.** High historic spend, no recent purchase — the highest-ROI win-back target in the base.
- **Demand is seasonal.** Revenue climbs from September and peaks at £1.16M in November, so retention campaigns should land before the Q4 ramp, not during it.

How it was built

Stage	Artifact	What it does
1. Clean	**data_cleaning.py**	Pandas pipeline: removes duplicates, null customer IDs, cancelled orders (`C`-prefixed invoices) and non-positive quantity/price, then derives `Total_Price`. 541,909 → 392,692 rows, with every exclusion logged and counted.
2. Model	**rfm_model.sql**	PostgreSQL CTEs and `NTILE(5)` window functions score each customer 1–5 on Recency, Frequency and Monetary value, concatenated into a three-digit RFM code and mapped to a business segment.
3. Visualise	**power_bi_dax.txt**	DAX measures (Total Revenue, Unique Customers, Average Order Value) and a `SWITCH(TRUE())` segmentation column, plus a matplotlib renderer for the static dashboard.

Modelling decisions worth noting

- **Recency is scored in reverse.** `NTILE(5) OVER (ORDER BY recency DESC)` puts the longest gap in quintile 1, so 5 always means 'bought most recently' — the direction every segment rule depends on.
- **The snapshot date is the last transaction plus one day**, so the most recent buyer scores a recency of 1 rather than 0 and no customer divides by zero.
- **Segment logic lives in SQL, not only in DAX**, so the label is identical in Power BI, in the static dashboard, and in any future tool reading the same table.



Left: revenue by Recency × Frequency quintile. Right: cumulative revenue by customer rank.

Recommended actions

Segment	Customers	Revenue	Recommended action
Champions	866	£5.66M (63.6%)	Early access, loyalty tier, referral asks. Protect first.
Loyal Customers	532	£1.07M (12.1%)	Cross-sell adjacent categories; raise basket size.
At Risk	532	£808K (9.1%)	Targeted win-back with a time-boxed incentive.
Cannot Lose Them	114	£218K (2.5%)	Personal outreach before Q4; highest ROI per contact.
New Customers	142	£32K (0.4%)	Onboarding sequence to drive a critical second order.
Lost	570	£126K (1.4%)	Low-cost automated reactivation only — no paid spend.

Where I would take it next

- **Measure, don't just segment.** Hold out a control group per campaign so win-back lift is attributable rather than assumed.
- **Score on a schedule.** Running the SQL nightly turns a one-off study into tracked movement between segments.
- **Add product affinity.** RFM says who to contact; basket analysis on the same table would say what to show them.